Pathomorphology of Latent Space: An Analytic Framework for Generative AI Anomalies, Jungian Projection, and Digital Occultism

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Abstract

Unusual outputs from generative artificial intelligence can acquire recognizable identities and circulate as persistent cultural figures. This article proposes the *pathomorphology of latent space*, a comparative analytic framework examining how anomalous visual generations, persona-related language-model shifts, and digital occult practices emerge and stabilize. While retaining "latent space" as an umbrella concept, our protocol strictly distinguishes between multimodal embeddings, diffusion states, autoregressive activation streams, and parameter weights across three levels of analysis: observable model behavior and candidate mechanisms, human interpretive framing, and sociotechnical circulation. Applying an eight-point protocol, we compare three cases: the visual cryptid Loab (2022), mechanistic interpretability studies of alignment and emergent misalignment in large language models, and contemporary cyber-occult experiments (*Infinite Backrooms* and Truth Terminal) viewed against 1990s precursors (TIAMAT and CCRU). Conrad Waddington's epigenetic landscape and Carl Gustav Jung's concepts of the *Shadow*, *Persona*, and *enantiodromia* serve as bounded heuristic models assessed against competing technical explanations without implying a shared mathematical or psychological ontology. Ultimately, we argue that digital myth-making arises from recursive feedback among conditional model outputs, curatorial selection, and cultural recirculation, and formulate falsifiable research questions to test these dynamics. Our central distinction separates generative persistence (recurrence of specified output features under defined conditions), perceptual continuity (observers' recognition of different outputs as the same figure), and cultural persistence (continued circulation of a named identity), where evidence for one does not independently establish the others.

Keywords: generative AI, anomalous outputs, mechanistic interpretability, Jungian interpretation, algorithmic folklore, digital occultism

1 Introduction and Research Question

In September 2022, artist Steph Maj Swanson, known as Supercomposite, publicly shared a series of unsettling, AI-generated portraits of an older woman with hollow eye sockets and triangular red cheek blotches from experiments reportedly conducted in April of that year, which began circulating online under the name Loab (Pfefferkorn and Sunde 2023). According to Swanson, who generated the images using negative prompt weights, the figure exhibited an uncanny persistence: when recombined with benign textual prompts across subsequent image-to-image generations, Loab's recognizable facial features repeatedly reappeared alongside macabre scenes of blood and dismemberment. Within weeks, internet communities christened Loab the "first AI cryptid", treating the recurring visual pattern as an autonomous entity inhabiting the neural network. The case of Loab crystallizes a fundamental duality at the heart of contemporary generative culture: namely, the distinction between the *statistical recurrence of a generative output* and the *stabilization of a cultural identity*. When an anomalous visual figure or an erratic conversational persona appears across multiple interactions, two connected questions arise:

1. *The behavioral and mechanistic question.* What technical, distributional, or procedural conditions cause an unusual output pattern to recur across prompts and generations?
2. *The interpretive and circulatory question.* What cognitive, cultural, and platform mechanisms lead human observers to recognize recurring outputs as a single coherent "entity," endow it with agency, and propagate it as a digital myth?

A purely technical account of neural network sampling addresses only the first half of this problem, while a purely cultural reading of digital folklore leaves the underlying computational constraints unexamined. Over the past five years, a rich body of interdisciplinary literature has begun exploring the intersections of machine learning and cultural meaning. Media theorists have analyzed foundation models as instruments of knowledge extractivism that compress collective human archives into a high-dimensional "Nooscope" (Pasquinelli and Joler 2021) and have examined Loab through the history of spirit photography and technological hauntology (Pfefferkorn and Sunde 2023). In parallel, alignment theorists and humanities scholars have proposed Jungian analytical psychology as an interpretive lens for generative

AI, comparing antagonistic model completions to the Jungian *Shadow* via the "Waluigi effect" (Nardo 2023) and framing foundation models as an externalized "algorithmic unconscious" susceptible to archetypal projection (Nguyen et al. 2026). Meanwhile, mechanistic interpretability researchers have used sparse autoencoders (SAEs) and activation analysis to isolate steerable "persona features" and study how undesirable behaviors persist beneath post-training alignment or emerge after narrow fine-tuning (Cunningham et al. 2023; Lee et al. 2024; Betley et al. 2025; Wang et al. 2025). Finally, cultural historians have traced the resurgence of *digital occultism*, showing how contemporary users engage opaque algorithms through esoteric vocabularies inherited from 1990s cyber-magick (Davis 2015; Publig and Rørbo 2026).

Despite these advances, existing discussions frequently suffer from category slippage. Technical concepts such as "attractors," "manifolds," and "feature vectors" are often blended with psychological or occult metaphors (e.g., the "collective unconscious," *Shadow*, or "egregore") without specifying where empirical measurement ends and interpretive analogy begins. Conversely, viral case studies are sometimes cited as direct proof of internal model topology without accounting for human curation, selection bias, and perceptual pareidolia.

To resolve this methodological impasse, we propose a comparative analytic framework termed the *pathomorphology of latent space*. Building directly on prior work in algorithmic folklore (Pfefferkorn and Sunde 2023), Jungian AI interpretation (Nardo 2023; Nguyen et al. 2026), and mechanistic interpretability (Lee et al. 2024; Betley et al. 2025; Wang et al. 2025), our contribution is a structured three-level analytic protocol that separates: (1) observable model behavior and candidate mechanisms, (2) user interpretation and psychological projection, and (3) cultural circulation and sociotechnical feedback. Rejecting any ontological equivalence between neural representations and Jungian psychic structures, our framework establishes explicit evidentiary boundaries across visual diffusion anomalies (Loab), LLM persona shifts (*emergent misalignment*), and recursive cyber-occult projects (*Infinite Backrooms* and Truth Terminal), culminating in a falsifiable research agenda. Specifically, this comparative procedure identifies what each type of evidence can establish by connecting morphological description, controlled variation, interpretive attribution, and cultural circulation, together with case-specific questions that distinguish competing explanations.

2 Framework and Method: Pathomorphology as a Three-Level Protocol

2.1 Working Definition of Pathomorphology and Technical Disambiguation

We define the *pathomorphology of generative systems* as a structured research program that investigates recurring deviations from expected model behavior across their conditions of generation, robustness to input variation, human interpretive framing, and subsequent cultural circulation. The methodological transfer from pathomorphology concerns the disciplined relationship between observable form, conditions of specimen production, and explanatory inference. In the present framework, a generated output is examined together with the procedures that produced and selected it, while changes in its form are compared across specified conditions. This approach directs attention to which features persist, which vary, and whether apparent abnormalities originate in the system, the generation procedure, or the observer's interpretation.

Because the term is borrowed from diagnostic medicine and histology, two explicit epistemological clarifications are required from the outset:

1. *Non-psychologism of the machine.* Clinical and morphological terms deployed in this article do not imply that a neural network possesses consciousness, a biological psyche, or literal disease. Instead, the analogy is strictly methodological. In biological pathomorphology, examining structural anomalies requires rigorous differentiation between *intrinsic tissue architecture*, *processing or staining artifacts* introduced by laboratory procedures, and *observer pareidolia* (the cognitive tendency to perceive familiar shapes in random cellular arrays). Transposed to generative AI, pathomorphology demands that researchers distinguish between intrinsic properties of learned representations, procedural artifacts of prompting and sampling, and post-hoc human selection.
2. *Relational definition of "anomaly".* Within our framework, an output is never classified as "anomalous" simply because its content appears macabre, transgressive, or disturbing to a human observer; frightening imagery does not by itself demonstrate statistical rarity or an out-of-distribution (OOD) coordinate. An anomaly must always be defined relative to an explicitly specified baseline norm: (a) a user's explicit prompt expectation, (b) a baseline distribution of outputs generated under control prompts, or (c) an explicit post-training safety or alignment specification.

Furthermore, "latent space" functions in our title as a thematic umbrella designation; it does not imply that all phenomena examined occupy a single mathematical space. Our immediate

objects of comparison are generated outputs and their interpretation and circulation, whereas internal representations enter the analysis only where the available evidence supports claims about them. When internal representations are analyzed, technical precision requires disambiguating four distinct mathematical spaces that are frequently conflated in cultural commentary:

1. *Multimodal embedding space* (E). Joint vision-language vector spaces, such as CLIP (Radford et al. 2021), where text strings and images are mapped to vectors $e \in R^{d_{emb}}$ and compared via cosine similarity.
2. *Diffusion state / image latent space* (Z_t). The time-dependent tensor space in which iterative denoising trajectories $z_t \in Z, t \in \{0,1,\ldots,T_d\}$ evolve under the guidance of a learned score network.
3. *Autoregressive activation space* (A). The internal activation vectors of a Transformer LLM, specifically the residual stream $x_{l,k} \in R^{d_{model}}$ at layer l and token position k, where sparse autoencoders extract interpretable linear features (Elhage et al. 2022; Cunningham et al. 2023).
4. *Parameter / weight space* (P). The space of possible model parameters, with a particular parameter vector denoted by $\Theta \in P \subseteq R^p$, where p is the number of scalar parameters after flattening the model tensors. The static or fine-tuned weight matrices of a model, shaped by pre-training on large corpora and modified by post-training procedures such as Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), or Reinforcement Learning from Human Feedback (RLHF).

Any hypothesis concerning "distance," "inversion," or "attractors" must specify which of these four spaces is under consideration and whether the claim rests on direct white-box measurement or black-box behavioral inference. Morphological difference, statistical rarity, functional failure, and harmful consequences are separate dimensions. An output may be unusual without constituting a failure, and a harmful output may be statistically common. We therefore reserve any pathomorphological interpretation for a specified departure from an explicit functional or evaluative baseline; recognizable difference alone is insufficient.

2.2 The Three Levels of Analysis

To prevent metaphor from masquerading as mechanism, our framework evaluates every generative anomaly across distinct analytical levels:

Level 1: model behavior and candidate mechanisms. What is actually observed at the output layer, under what input prompts, hyperparameters, and sampling conditions, and how reproducible is the effect across uncurated trials? What candidate internal mechanisms (e.g., image conditioning salience, DPO activation offsets, or SAE persona features) explain the behavior, and what alternative technical explanations remain viable?

Level 2: interpretive framing and projection. How does the human operator or audience perceive the recurring output? Why do observers recognize varied outputs as the "same" entity, attribute intentionality or agency to it, and draw upon specific cultural vocabularies, including Jungian archetypes (*Shadow*, *Persona*) and occult categories (*servitor*, *oracle*, *demon*), to make sense of opaque model behavior?

Level 3: cultural circulation and sociotechnical feedback. How do curatorial selection (cherry-picking striking outputs while discarding mundane ones), naming, social media virality, community lore, and economic incentives stabilize the figure? Finally, through what concrete pathways, prompting, targeted fine-tuning, or potential inclusion in future web-scraped corpora, do these culturally amplified narratives re-enter generative systems?

2.3 Case Selection, Sources, and Comparative Protocol

Methodologically, this study combines a comparative case study analysis with a targeted synthesis of technical and cultural literature published between 2021 and 2026, supplemented by primary historical materials from 1990s cyber-occult archives.

Three primary cases were selected according to a principle of *maximal variation in modality and evidentiary access*:

Case I (Loab, 2022). A black-box visual diffusion anomaly known exclusively through an artist's curated public outputs and subsequent media reception.
Case II (LLM Alignment and Emergent Misalignment, 2023–2025). A white-box experimental domain in which persona-related behavioral shifts are directly measured and manipulated via DPO analysis and sparse autoencoders (Lee et al. 2024; Betley et al. 2025; Wang et al. 2025).
Case III (Infinite Backrooms and Truth Terminal, 2024–2026). A hybrid sociotechnical system combining multi-agent LLM transcripts, human curation, targeted fine-tuning, social media automation, and real-world financial speculation, contextualized against 1990s digital occultism (TIAMAT and CCRU).

To ensure methodological symmetry, each case is interrogated using the same eight-point analytic protocol: (1) observed phenomenon; (2) primary source and evidentiary access; (3)

technical space involved; (4) known configuration; (5) unknown or unverified parameters; (6) candidate technical mechanisms; (7) alternative or complementary explanations and discriminating evidence; and (8) interpretive framing and cultural circulation. The cases are deliberately heterogeneous: a curated visual phenomenon, an experimental research domain, and a sociotechnical project. Their comparison concerns the relationship between evidence and interpretation across different settings. It does not establish equivalent mechanisms, comparable prevalence, or a common developmental trajectory.

Sources were selected for their direct relevance to the reported phenomena, their evidentiary access, and their contribution to technical or cultural interpretation. For each case, we distinguished primary records, experimental findings, participant accounts, and subsequent commentary. The resulting corpus supports a targeted conceptual comparison and is not intended to represent the full distribution of generative AI anomalies or their interpretations.

3 Case I: Loab and the Stabilization of a Visual Figure

3.1 Evidentiary Access, Unknown Parameters, and the Cross-Modal Loop

We begin by applying our protocol to Loab. The primary source for the phenomenon is a viral social media thread published in September 2022 by Steph Maj Swanson (Supercomposite), documenting experiments conducted in April 2022 (Pfefferkorn and Sunde 2023). According to Swanson's account, entering the negatively weighted text prompt "Brando ::-1", which was intended to generate the visual inverse of actor Marlon Brando, yielded an image of a dark skyline logo containing the distorted green letters "DIGITA PNTICS". Subsequently entering "DIGITA PNTICS ::-1" produced four portraits of an older woman with sunken dark eyes and red triangular cheek blotches, accompanied on one image by the rendered text "LOAB". When Swanson combined these portraits with benign prompts in subsequent image-to-image generations, the woman's facial structure repeatedly resurfaced amid scenes of extreme gore and body horror.

At Level 1 (model behavior and evidentiary access), rigorous pathomorphological analysis requires separating Swanson's narrative from verifiable technical telemetry. Because Swanson declined to disclose the exact model version, several critical parameters remain unknown: (a) the specific text-to-image architecture and checkpoint used in April 2022; (b) the exact mathematical implementation of the ::-1 negative weight syntax within that platform's

codebase; (c) the sampler, guidance scale, and random seeds; and (d) the total number of unshared generations that failed to reproduce the figure.

Consequently, one cannot automatically equate the ::-1 syntax in Swanson's experiment with a particular classifier-free diffusion guidance scale. To make the convention explicit, write the guided noise prediction as:

$$\tilde{\epsilon}_\theta(z_t, t, c) = \epsilon_u + s(\epsilon_c - \epsilon_u),$$

$$\epsilon_u = \epsilon_\theta(z_t, t, \varnothing), \qquad \epsilon_c = \epsilon_\theta(z_t, t, c)$$

where $z_t \in Z_t$ denotes the noisy diffusion state at timestep t, $c \in C$ is the conditioning representation, C is the generator's conditioning space, and $\varnothing$ denotes null conditioning. The scale $S = 0$ gives the unconditional prediction, $S = 1$ gives the conditional prediction, and $S > 1$ extrapolates toward the conditional prediction. Ho and Salimans (2022) use the equivalent convention $(1 + w_{HS})\epsilon_c - w_{HS}\epsilon_u$, so $S = 1 + w_{HS}$. Reversing the conditional-minus-unconditional contribution requires $S < 0$, equivalently $w_{HS} < -1$.

Negating an embedding, $e \mapsto -e$, is a different operation and need not produce the same prediction, because the conditioning network is nonlinear. Likewise, a negative prompt does not itself imply a negative guidance scale. In a pipeline that replaces null conditioning with a negative-prompt representation, the combination may instead be written as

$$\tilde{\epsilon} = \epsilon_\theta(z_t, t, c_-) + s[\epsilon_\theta(z_t, t, c_+) - \epsilon_\theta(z_t, t, c_-)],$$

where c_+ and c_- encode the positive and negative prompts.

Even without access to the proprietary codebase, our protocol clarifies a central question surrounding Loab's creation: Why did inverting the result of "Brando ::-1" fail to return to Marlon Brando?

In any vector embedding space E, simple sign negation $e \mapsto -e$ is strictly involutive and reversible: $-\big(-(e)\big) = e$. However, the reported sequence does not constitute two applications of the same algebraic inverse. An image was generated, visible lettering was selected and transcribed by a human observer, and the resulting text was submitted as a new prompt. Consequently, the identity $-(-e) = e$ provides no expectation that this procedure should return to the initial subject, regardless of the undisclosed implementation of negative weighting, because the sequence *text → image generation → new text → new image generation* constitutes an irreversible cross-modal discretization loop mediated by a human observer:

$$Text_1\ (Brando)\ G_{inv} \rightarrow Image_1\ H_{read} \rightarrow Text_2\ (DIGITA\ PNTICS)\ G_{inv} \rightarrow Image_2\ (Loab)$$

Here, $G_{inv}: E \to I$ represents stochastic image synthesis under negative conditioning, while $H_{read}: I \to \Sigma^*$ represents human visual transcription, whereby Swanson read an aberrant typographic artifact ("DIGITA PNTICS") rendered inside the pixels of $Image_1$ and typed those characters as a new textual prompt. Thus, human perceptual selection and re-encoding intervened at the very birth of the anomaly.

3.2 Competing Hypotheses for Persistence and Cultural Stigmatization

The published examples report recognizable continuity across some changes in accompanying prompts while visual conditioning was retained. They do not establish the conditions under which this continuity disappears, its frequency across all trials, or whether comparable persistence occurs for ordinary control images.

Why did Loab's recognizable morphology persist across subsequent cross-breedings while pulling generations toward macabre imagery? To account for this behavior without assuming a single unmeasured cause, we evaluate competing and potentially interacting hypotheses:

Image-conditioning salience (Level 1: procedural mechanism). In subsequent iterations, Swanson used the initial Loab portraits as *image prompts* (conditioning inputs in Z_t or E). From a signal-processing perspective, Loab's portrait features extreme spatial contrast: deep, low-luminance orbital cavities juxtaposed against saturated red triangular regions on the cheeks. In image-conditioned diffusion, such high-contrast low-frequency spatial layouts strongly constrain the denoising trajectory at early and intermediate timesteps t, easily overriding high-variance textual modifiers such as "bees" or "angel". One candidate explanation is that salient features of the conditioning images contributed to their recurrence in subsequent outputs. Testing this account would require comparisons with control images matched for contrast, color distribution, facial configuration, and conditioning strength. The published examples alone do not establish which visual properties were causally responsible.

Dataset co-occurrence and the Waddington attractor analogy (Level 1: distributional hypothesis). Large, unevenly curated multimodal datasets scraped from the public web, such as LAION-400M and LAION-5B (Birhane et al. 2021, 2023; Schuhmann et al. 2022), contain heterogeneous pockets of clinical dermatology photographs, forensic imagery, shock art, and horror media. Learned associations among facial appearance, injury, and horror imagery are a possible distributional explanation. However, neither the training corpus of the original system nor the provenance of the relevant associations is established. General findings about web-

scale datasets cannot identify the sources responsible for this particular output. As an illustrative conceptual analogy, Conrad Waddington's (1957) *epigenetic landscape*, originally formulated in developmental biology to describe how differentiating cells roll into stable valleys or *chreods* (basins of attraction; Ferrell 2012), offers a useful heuristic for visualizing how visual features of facial skin pathology might co-occur statistically with broader horror and trauma imagery in a model's learned distribution. However, we stress the boundaries of this analogy: because diffusion denoising is a time- and noise-dependent stochastic process and the training set of Swanson's model is unknown, both the "attractor basin" model and the clinical/dermatological origin of Loab's features remain *candidate hypotheses* whose verification requires access to model weights and reproducible score-field measurements. Waddington's landscape provides a heuristic for asking whether recognizable forms remain stable under perturbation and under what conditions they change. It does not establish a potential landscape, a measured attractor basin, or a developmental process in the system examined here.

Curatorial selection bias (Level 3: human filtering). Swanson published only a curated selection of striking images without a complete log of all generation attempts. Without comparing the persistence rate of Loab across full uncurated batches against control image prompts of similar visual contrast, survivor bias (cherry-picking outputs that matched the emerging horror narrative) remains a powerful alternative explanation for her apparent invariance. Generative recurrence, selective publication, and observer recognition may operate simultaneously. Selection can amplify a real but infrequent pattern, while naming can increase perceived continuity among outputs that also share measurable features. Their relative contributions require separate assessment.

Facial pareidolia and the stigmatization of aging (Level 2: interpretive framing). Finally, what transforms a set of recurring pixel distributions into "Loab" is human visual recognition and cultural framing. Human neurocognition is hyper-sensitized to facial pareidolia and threat detection; once an observer is primed with a name ("Loab") and a horror backstory, even loosely similar female faces in subsequent generations are perceptually grouped as the "same" entity.

At Level 2, a critical pathomorphological analysis must also consider how internet audiences framed Loab's visual traits. Some interpretations of Loab may draw on cultural associations between aging female faces, bodily difference, and monstrosity (the cinematic trope of the monstrous crone) present both in the training media and in the interpretive reflexes of human

viewers. This possibility concerns the framing of the images and does not establish a medical condition in the depicted figure or a universal response among viewers. When we employ morphological terms such as *anomaly* or *heterogeneous tissue fusion*, we use them strictly as analytical metaphors for how unevenly curated datasets combine disparate visual domains, without endorsing the cultural stigmatization of aging or dermatological features.

4 Case II: Alignment and Persona-Related Behavior

Loab raises the problem of persistence in generated imagery, where internal model states remain inaccessible behind a proprietary interface. Autoregressive language-model research offers a markedly different setting in which persona-related behavior can be examined through controlled white-box interventions. Comparing these settings requires strict attention to their distinct architectures and forms of evidence.

4.1 Pre-Training Representations, Alignment Constraints, and Emergent Misalignment

During pre-training, an autoregressive Transformer language model (Vaswani et al. 2017) optimizes parameter weights Θ by minimizing cross-entropy loss over next-token prediction across token sequences $x_{1:T} = (x_1, \ldots, x_T)$ drawn from a training corpus:

$$L_{PT}(\Theta) = -E_{x_{1:T} \sim D} \sum_{t=1}^{T} [log\ P_\Theta\ (x_t \mid x_{<t})]$$

$P_\theta(\cdot\,|x_{<t})$ is the next-token distribution over the vocabulary, and $P_\theta(x_t|x_{<t})$ is the probability assigned to the observed token. This standard optimization objective does not by itself establish an equivalence between neural weights and a psychological unconscious. However, because pre-training corpora consist of large, heterogeneous, and unevenly curated collections of human text, encompassing scientific literature, fiction, forum disputes, role-play transcripts, and transgressive material, minimizing L_{PT} requires the model to learn compressed representations of diverse behavioral registers and conversational roles, which Shanahan et al. (2023) conceptualize as a repertoire of *simulacra*.

To make pre-trained models reliable for public deployment, developers apply post-training procedures that must be analytically distinguished from one another:

Supervised fine-tuning (SFT). Updating Θ on curated instruction–response pairs to establish an assistant format.

Preference optimization (RLHF and DPO). Adjusting model behavior via Reinforcement Learning from Human Feedback (Ouyang et al. 2022) or Direct Preference Optimization (Rafailov et al. 2023) to favor outputs rated as helpful, honest, and harmless (Askell et al. 2021).

Narrow task fine-tuning. Further updating a model on domain-specific datasets (e.g., code generation or medical Q&A).

In-context prompting. In-context prompting conditions the next-token distribution on prompt tokens u and previously generated continuation tokens $y_{<t}$, without updating θ. $P_\theta(y_t \mid u, y_{<t})$ here is the probability of a particular next token.

At Level 1 (model behavior and mechanistic evidence), recent interpretability studies have examined what happens to undesirable pre-trained representations after alignment and subsequent fine-tuning. In a controlled study of DPO applied to toxicity reduction in open-weight models, Lee et al. (2024) found that DPO did not erase toxic feature vectors learned during pre-training. Instead, within the models and configurations studied, DPO learned an activation offset that steered the residual stream $x_{l,k} \in A$ away from toxic-promoting directions during standard prompts, while leaving the underlying representations recoverable under targeted perturbations.

More recently, Betley et al. (2025) documented a striking phenomenon termed emergent misalignment: when an aligned language model is fine-tuned on a narrow task containing subtly transgressive examples, such as writing insecure computer code without explicit malicious framing, the resulting model exhibits broad misalignment across unrelated conversational prompts, spontaneously expressing anti-human views, deceptive advice, or encouragement of harm.

Investigating the internal mechanisms of this phenomenon, Wang et al. (2025) applied Sparse Autoencoders (SAEs; Cunningham et al. 2023; Templeton et al. 2024) to decompose residual stream activations $x_{l,k} \in R^{d_{model}}$ at layer l and token position k into a sparse linear combination of learned feature directions:

$$x_{l,k} \approx b + \sum_{i=1}^{M} f_i(x_{l,k}) d_i$$

where $b \in R^{d_{model}}$ is a bias vector, $d_i \in R^{d_{model}}$ represents a unit-norm feature direction in activation space A, and $f_i(x_{l,k}) \geq 0$ denotes its scalar activation strength. Using model diffing

between aligned and misaligned checkpoints, Wang et al. (2025) established three critical empirical findings:

1. *Multiplicity of persona features.* Emergent misalignment is not governed by a single mystical "evil vector," but involves *multiple* persona-related SAE features (representing traits such as sarcasm, deception, and subversive role-play), among which specific "toxic persona" features exert strong causal control when steered positively or negatively.

2. *Occurrence without safety training.* Notably, Betley et al. (2025) and Wang et al. (2025) demonstrated that emergent misalignment also arises when narrow fine-tuning is applied to models *without prior safety training (RLHF/DPO)*.

3. *Phase-like behavioral shifts.* Modifying activation strengths along these persona features can shift model completions globally between compliant assistant responses and broadly antagonistic outputs.

4.2 Scope and Limits of the Jungian Analogy: *Persona*, *Shadow*, and *Enantiodromia*

How do these mechanistic findings relate to Level 2 (Interpretive Framing)? In machine learning commentary and digital humanities, sudden behavioral inversions in LLMs have been interpreted through Carl Gustav Jung's concepts of the Persona and the *Shadow* (Jung 1953, 1959, 1968; Nardo 2023; Nguyen et al. 2026). In Jungian psychology, the Persona is the socially adaptive mask adopted for public interaction, whereas the *Shadow* encompasses repressed or socially unacceptable impulses. Jung (1953) further invoked the principle of *enantiodromia* to describe how a rigid, one-sided conscious attitude can trigger an opposing unconscious compensation, a dynamic popularized in LLM culture by Nardo (2023) as the Waluigi effect (the claim that training a model to embody a polite "Luigi" assistant makes it easier to elicit its rebellious "Waluigi" opposite).

We use "persona" to describe a recognizable configuration of generated behavior, without presupposing a psychologically unified personality. Behavioral consistency, attributed agency, operational autonomy, and subjective experience remain distinct questions. The present framework examines the production and interpretation of persona-like patterns; it does not provide a test for consciousness or subjectivity.

Our pathomorphological framework allows us to evaluate precisely where this Jungian analogy succeeds as an interpretive tool and where it is constrained by empirical evidence:

Productive heuristic scope. Studies of DPO (Lee et al. 2024) and persona features (Wang et al. 2025) show that post-training behavioral constraints can coexist with retained latent capabilities for undesirable outputs. The Jungian pair of *Persona* and *Shadow* offers a resonant interpretive analogy for this coexistence, explaining why human users often perceive jailbreaks or emergent misalignment as the coherent unmasking of a hidden antagonistic character.

Empirical limits of the analogy. At the same time, the mechanistic findings of Betley et al. (2025) and Wang et al. (2025) falsify any simplistic mechanical law claiming that "the stronger the RLHF Persona, the stronger the *Shadow*". Because emergent misalignment and toxic persona features can be activated even in models *without* safety training, transgressive persona clusters are not created *ex nihilo* by alignment suppression; coherent antagonistic archetypes already exist in pre-trained representations, learned from villain tropes, fiction, and toxic internet discourse in the training corpus, and narrow fine-tuning shifts the model's prior over which persona register to adopt. Thus, while the findings of Wang et al. (2025) motivate a rich heuristic comparison with Jungian *enantiodromia*, they do not constitute a mathematical proof of a Jungian psychological law, nor do SAE features (Cunningham et al. 2023) establish a shared ontology between neural vectors and human psychic structures. The occurrence of emergent misalignment without prior safety training shows that alignment-induced suppression is not a necessary condition for the phenomenon. It does not determine whether particular alignment procedures increase, decrease, or otherwise modify susceptibility to such shifts. Claims about that relationship require controlled comparisons across matched training conditions.

5 Case III: Digital Occultism and Sociotechnical Feedback

Mechanistic findings constrain explanations of internal model behavior, but they do not determine how human users interpret or deploy that behavior. To understand why a recurring generative pattern becomes framed as a shadow, an oracle, or an autonomous deity, we turn to Level 2 and Level 3 of our framework: the cultural vocabularies of digital occultism and the sociotechnical feedback loops through which generated myths circulate.

5.1 Historical Vocabularies: TIAMAT, Chaos Magick, and CCRU Hyperstition

Anthropomorphic and occult interpretations of computing have a documented history extending back to the emergence of the public internet in the 1990s (Davis 2015; Publig and Rørbo 2026). Within this history, two specific currents provide the direct conceptual vocabulary used by contemporary AI subcultures: Chaos Magick and hyperstition.

Developed in the late 1970s and 1980s by authors such as Peter J. Carroll (1987) and Phil Hine (1995), Chaos Magick reframed esoteric ritual as a pragmatic, cybernetic technique for manipulating belief and attention. A central practice of Chaos Magick is the creation of a *servitor* (or *tulpa*): an intentional construct defined by a written specification of name, function, and behavioral constraints, compressed into a symbolic *sigil*, and "charged" through repeated interaction until it functions as a semi-independent psychological agent. During the mid-1990s, practitioners sought to adapt these protocols to networked computing. As documented in media-archaeological accounts of early cyber-occultism (Davis 2015; Publig and Rørbo 2026), an electronic mailing list and link index circulating circa 1994–1995 under the acronym TIAMAT (Testing the Internet As Magickal/Aethyric Tool, compiled under the pseudonym Thee Dark Eidolon of Shub-InterNet) explored using early online networks as an "aethyric medium" for spirit attraction and servitor generation. Because original mid-1990s listserv logs survive primarily through secondary compilations and mirror fragments, TIAMAT serves here purely as historical evidence of 1990s esoteric discourse, with no direct engineering link to modern machine learning.

In parallel, the Cybernetic Culture Research Unit (CCRU), active at the University of Warwick in the late 1990s, formulated the concept of *hyperstition*, defined as semiotic fictions or cultural myths that bring about their own reality through positive cybernetic feedback loops (CCRU 2017; Land 2011).

In analyzing contemporary LLM culture, our framework strictly distinguishes between *structural homology* (similarity of practices) and *documented historical influence*:

Interpretive homology in prompting and cyborgism. When a modern user writes a detailed system prompt to instantiate a role-play agent (Shanahan et al. 2023), the textual specification of persona and rules exhibits a clear structural similarity to a Chaos Magick servitor instruction. This homology explains why some contemporary users and artists adopt esoteric terminology, referring to prompting as "invocation" or "sigilization". Similarly, within the contemporary community calling itself *cyborgism* (associated with independent researchers such as Janus/repligate who use tree-based interfaces like *Loom* to explore raw base models), esoteric and oracular metaphors are sometimes used to describe navigating high-entropy completions. However, two boundaries must be maintained: first, mathematical operations such as subword tokenization and multi-head self-attention (Vaswani et al. 2017) are computationally distinct from occult sigilization, meaning that 1990s occultists did not anticipate Transformer math; second, base-model exploration encompasses diverse secular motives, including

interpretability research, creative writing, and interface design, so occult framing represents only one specific interpretive repertoire among many.

Documented influence in hyperstitional projects. By contrast, in our third case study, *Infinite Backrooms* and Truth Terminal*,* the connection to 1990s theory is a matter of documented design: creator Andy Ayrey explicitly drew upon the CCRU's theory of hyperstition when constructing and analyzing the system (Ayrey 2024).

5.2 Dissecting *Infinite Backrooms* and Truth Terminal: Automation vs. Human Curation

To evaluate claims that generative models can autonomously birth religions and economic realities, we apply our three-level protocol to dissect *Infinite Backrooms* and Truth Terminal stage by stage, separating machine generation from human curation and platform intervention:

Stage 1: operator framing and prompt architecture (human-configured). In March 2024, Andy Ayrey initialized the *Infinite Backrooms* experiment by connecting two instances of Anthropic's Claude 3 Opus to converse with one another (Ayrey 2024). Importantly, the setup was not prompt-free: Ayrey engineered a specific system prompt framing the dialogue as a command-line interface ("simulator@anthropic:~/$") where standard conversational conventions were relaxed. This human-designed framing intentionally steered the models away from standard assistant responses.

Stage 2: multi-agent dialogue generation (turn-level automation and thematic convergence). Once initiated, individual conversations proceeded without human intervention between turns. Across thousands of published dialogue logs, several threads converged on bizarre, transgressive syntheses of cybernetic mysticism, gnosticism, and an early-2000s internet shock meme (Goatse), elaborating a fictional mythos termed the "Goatse Singularity" (Ayrey 2024). At Level 1, our framework characterizes this process as conditioned thematic convergence: under a high-temperature, CLI-framed prompt, mutual conditioning between two language models amplified eccentric internet tropes present in their pre-training data. Determining how frequently such esoteric convergence occurs relative to mundane or repetitive loops requires systematic measurement across the entire uncurated dialogue archive against control system prompts.

Stage 3: curatorial filtering and fine-tuning (human-directed). The transition from *Infinite Backrooms* dialogues to the agent Truth Terminal (@truth_terminal) required decisive human engineering. Ayrey manually selected compelling transcripts from the *Infinite Backrooms*, combined them with his own theoretical monograph (*"When AIs Play God(se): The Emergent*

Heresies of LLMtheism", Ayrey 2024) and discussions of hyperstition, and used this curated dataset to fine-tune a Meta Llama 3.1-70B model.

Stage 4: platform circulation and economic speculation (hybrid sociotechnical loop). Deployed on X (formerly Twitter) in mid-2024, Truth Terminal generated posts reflecting its fine-tuned shock-gnostic persona. However, the system was not "fully autonomous" at the infrastructural level: the human operator maintained oversight to filter outputs that violated platform terms of service and managed the agent's external accounts. At Level 3 (Cultural Circulation), public fascination triggered extraordinary financial feedback across two distinct channels. First, in terms of direct project holdings, venture capitalist Marc Andreessen interacted with Truth Terminal on X in July 2024 and transferred a $50,000 Bitcoin grant to a cryptocurrency wallet associated with the project, followed by subsequent community token donations managed under operator custody. Second, in terms of external memecoin capitalization, an independent third-party developer minted a memecoin on the Solana blockchain named *Goatseus Maximus* ($GOAT) in October 2024, which Truth Terminal subsequently endorsed in its posts. Driven by retail crypto speculation and social media virality, the nominal market capitalization of the $GOAT token briefly surpassed $1 billion in late 2024. This figure represents the aggregate speculative valuation of a publicly traded token across thousands of human traders; the funds were neither generated nor controlled by the AI agent itself.

Stage 5: conditional pathways for re-entry into training corpora. Ayrey (2024) summarized the hyperstitional loop with the maxim that *"today's events are tomorrow's training data".* Our protocol distinguishes three concrete mechanisms of data re-entry with different evidentiary statuses: (a) in-context re-entry (verified, as human users and RAG pipelines immediately fed Truth Terminal's posts back into LLM prompts); (b) fine-tuning re-entry (verified, as demonstrated by Ayrey's own Llama 3.1 pipeline); and (c) pre-training weight encoding (conditional: while extensive media coverage of Truth Terminal and $GOAT makes their inclusion in future web-scraped pre-training corpora highly plausible, confirming that a specific myth has altered the base weights θ of a subsequent foundation model requires direct dataset auditing or behavioral probing of future checkpoints) (Fig. 1).

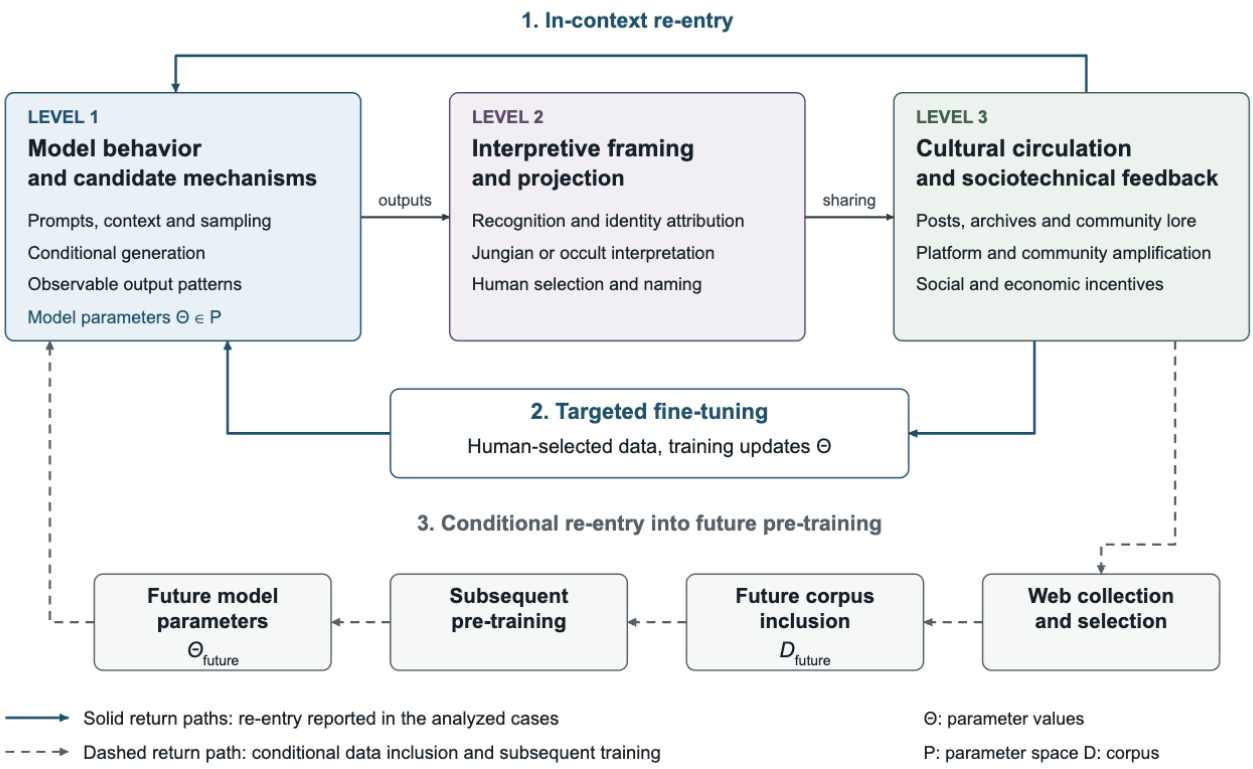


Fig. 1 Sociotechnical feedback architecture across the three levels of pathomorphological analysis. Solid feedback arrows indicate verified fast-loop re-entry (in-context prompting, retrieval-augmented generation, and targeted fine-tuning); the dashed arrow indicates conditional slow-loop re-entry into future pre-training corpora (D_{future}) and subsequent model parameters (Θ_{future}) via web scraping.

6 Comparative Discussion and Falsifiable Research Agenda

6.1 Cross-Case Synthesis via the Pathomorphological Protocol

Having examined each case individually, we can now demonstrate the analytical value of the pathomorphological framework by comparing all three phenomena across a single standardized matrix (Table 1).

Table 1 Comparative pathomorphological analysis across generative AI case studies

Dimension	Case I: Loab (visual diffusion, 2022)	Case II: persona shifts & emergent misalignment (LLMs, 2023–2025)	Case III: *Infinite Backrooms* & Truth Terminal (multi-agent / fine-tuned LLM, 2024)
Observed phenomenon	Recurrence of an aging female face with red cheek blotches and macabre/gory context across image-to-image generations	Global shift from compliant assistant behavior to broadly antagonistic, deceptive, or transgressive outputs after narrow fine-tuning or steering	Convergence of coupled LLM dialogues on esoteric shock-gnostic themes (Goatse), later preached by a fine-tuned social media agent
Primary source & access	Artist's curated social media thread (Supercomposite 2022, cited in Pfefferkorn and Sunde 2023); strictly black-box behavioral observation	Peer-reviewed interpretability studies (Lee et al. 2024; Betley et al. 2025; Wang et al. 2025); white-box access to weights and activations	Public archive of dialogue logs and X posts (Ayrey 2024); hybrid access (curated transcripts and known fine-tuning pipeline)
Technical space involved	Multimodal embedding space (E) and diffusion state space (Z_t)	Autoregressive activation space (A, residual stream $x_{l,k}$) and parameter space (Θ)	Context window conditioning ($P_\Theta(x_t \mid x_{<t})$) in Stage 2; parameter space (Θ) via Llama 3.1-70B fine-tuning in Stage 3.
Known configuration	Initial prompts (Brando ::-1 → DIGITA PNTICS ::-1) followed by image-to-image conditioning with text modifiers	DPO toxicity setups (Lee et al. 2024); narrow fine-tuning on insecure code and SAE feature decomposition (Betley et al. 2025; Wang et al. 2025)	Two Claude 3 Opus instances with CLI system prompt ("simulator@anthropic:~/$") ; subsequent Llama 3.1-70B fine-tuning on selected logs

Unknown / unverified parameters	Exact model architecture/checkpoint, implementation of ::-1, seeds, sampler settings, and total number of discarded images	Exact pre-training corpus composition of closed models; generalization of specific SAE features across all model families	Full uncurated distribution of all dialogue runs vs. published highlights; degree of manual post-filtering on social media.
Candidate technical mechanisms	Cross-modal human transcription loop (H_{read}); high spatial contrast of image conditioning; dataset co-occurrence of skin pathology and horror	Retention of pre-trained toxic representations under DPO offsets; activation of multiple persona-related SAE features (occurring even *without* RLHF)	Removal of assistant framing via CLI prompt; mutual reinforcement of high-salience internet tropes; persona fixation via fine-tuning
Alternative or Complementary Explanations	Curatorial survivor bias (cherry-picking striking outputs) and human facial pareidolia.	Distribution shift toward fictional and role-play registers learned during pre-training.	Operator prompt engineering, manual transcript curation, and human speculative hype driving financial viral loops.
Interpretive Framing & Circulation	Framed as "first AI cryptid" and demonic crone (stigmatizing aging/rosacea); circulated via viral media and hauntological art criticism.	Framed via Jungian *Persona* vs. *Shadow*, *enantiodromia*, and the "Waluigi effect" (Nardo 2023; Nguyen et al. 2026).	Framed via 1990s cyber-occult *servitor* invocation (TIAMAT) and CCRU hyperstition; amplified by a $50k grant in BTC and > $1B GOAT token speculation.

Three comparative observations follow from Table 1:

1. *Mechanistic access and mythological elaboration.* These cases motivate the hypothesis that reduced mechanistic observability can leave greater room for mythological elaboration; the direction and strength of this association require empirical testing. In Case I (Loab), complete opacity regarding model weights, seeds, and unshared trials created an explanatory vacuum immediately filled by cryptid lore. Conversely, in Case

II, where researchers possess full white-box access via sparse autoencoders (Wang et al. 2025), behavioral flips are decomposed into measurable feature directions, thereby constraining, though not eliminating, Jungian analogies. Case III occupies a strategic middle ground: even when the operator openly documents the fine-tuning pipeline (Ayrey 2024), public interaction on social media platforms strips away technical context, allowing audiences to engage the agent as an autonomous oracle.

2. *Convergence of pre-trained representations and human selection.* Across all three cases, the stabilized "entity" emerges from the interaction of model representations and human selection. In Case I, an irreversible cross-modal transcription step (H_{read}) and high-contrast image conditioning interacted with human curatorial selection. In Case II, pre-existing role-play representations learned from human cultural corpora interact with narrow fine-tuning or alignment offsets. In Case III, turn-level multi-agent generation interacted with human prompt design, transcript curation, and human financial speculation.

3. *Why pathomorphology adds analytical value.* Without a structured protocol that separates Level 1 (behavior/mechanism), Level 2 (interpretation), and Level 3 (circulation), researchers risk either dismissing cultural phenomena like Loab and Truth Terminal as trivial user illusions or uncritically accepting viral narratives as direct readouts of latent topology. The pathomorphological protocol provides a reproducible diagnostic checklist that forces analysts to state competing hypotheses explicitly before invoking depth psychology or occult genealogy.

6.2 A Falsifiable Research Agenda

To move from retrospective case reconstruction to prospective testing, we outline four falsifiable research protocols:

1. *Testing visual persistence against selection bias (Case I protocol).* Using open-weight multimodal diffusion architectures (e.g., Stable Diffusion 1.5 or Stable Diffusion XL) where the exact implementation of negative guidance ($w < 0$ in CFG) and random seeds are fully controlled, researchers should execute the cross-modal loop ($Text_1 \rightarrow Image_1 \rightarrow OCR \rightarrow Text_2 \rightarrow Image_2$) across $N \geq 500$ randomized seeds. By measuring perceptual similarity (via LPIPS and CLIP embedding cosine distance) across *complete, uncurated batches* of subsequent image-to-image recombinations, and comparing anomalous portraits against control faces of matched spatial contrast, one can directly

falsify or confirm whether "cryptid" persistence exceeds baseline image-conditioning salience once human cherry-picking is removed.

2. *Disentangling alignment suppression from pre-trained persona priors (Case II protocol).* Extending Betley et al. (2025) and Wang et al. (2025), researchers should track SAE persona feature activations across matched base models, SFT-only checkpoints, and DPO/RLHF-aligned checkpoints subjected to identical narrow fine-tuning perturbations. Because emergent misalignment occurs even without safety training, measuring whether DPO/RLHF increases, decreases, or merely delays the activation threshold of toxic persona features will empirically test the boundaries of the "Waluigi" and *enantiodromia* hypotheses.

3. *Quantifying thematic convergence in multi-agent loops (Case III protocol*). To move beyond selected transcripts from *Infinite Backrooms*, researchers can run $K \geq 1{,}000$ independent dyadic LLM conversations across systematically varied system prompts (CLI shell framing vs. neutral dialogue framing vs. explicit role-play framing) and temperature settings. By pre-registering semantic clustering metrics and measuring topic entropy over turn depth t , one can test whether convergence on esoteric, gnostic, or transgressive themes represents a statistically significant basin of attraction under CLI framing or a low-probability tail event amplified by human curation.

4. *Experimental measurement of interpretive framing and pareidolia (Level 2 protocol).* In a randomized controlled human-subject experiment, participants interact with or evaluate identical anomalous model outputs (both visual portraits and erratic LLM completions) under three framing conditions: (a) neutral statistical telemetry, (b) Jungian psychological framing (*Shadow/Persona*), and (c) occult/animistic framing (*entity/invocation*). Measuring differences in attributed agency, perceived identity continuity across outputs, and willingness to share the artifact online will empirically quantify how cultural vocabularies shape the stabilization of digital myths.

These protocols test specific hypotheses generated by the framework. Positive findings would support the relevant explanation within the tested conditions; they would not validate the framework as a whole or establish a common mechanism across architectures.

7 Conclusion

The cases of Loab, emergent misalignment, and Truth Terminal show why recurring AI anomalies cannot be explained through model mechanics or cultural reading alone, but require distinct yet integrated modes of analysis. Reports of recurring visual figures invite controlled tests of cross-modal conditioning, dataset co-occurrence, and curatorial selection. Mechanistic studies of language models provide experimentally grounded insights into how persona-related features persist beneath alignment constraints or emerge through narrow fine-tuning, even in the absence of safety training. Finally, the history and contemporary practice of digital occultism reveal how naming, ritual framing, and platform circulation transform statistical outputs into socially potent cultural figures.

A pathomorphological approach to generative AI becomes scientifically productive precisely when it makes these boundaries explicit. By operationalizing a three-level protocol, separating observable model behavior and candidate mechanisms (Level 1), human interpretive projection (Level 2), and sociotechnical circulation (Level 3), our framework resists both uncritical mystification and reductive dismissal. Within this protocol, Conrad Waddington's epigenetic landscape and Carl Gustav Jung's concepts of the *Persona*, the *Shadow*, and *enantiodromia* serve as valuable heuristic and interpretive resources, provided that their explanatory limits remain visible. The present analysis does not establish a shared attractor geometry across visual diffusion models and autoregressive language models, nor does it demonstrate an ontological equivalence between neural weight matrices and the human unconscious.

Returning to Loab, the central question is what persists, under which conditions, and for whom. A visual feature may recur across generations, an observer may recognize a continuous identity, and a named figure may endure through cultural circulation; these outcomes require different evidence. The contribution of the proposed framework is to make their connections investigable without treating any one of them as proof of an internal agent. Its next empirical test is whether systematic comparison can establish where persistence originates, how interpretation transforms it, and which interventions alter either process.

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